

Garvit Joshi

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EDUCATION

R.V. College of Engineering

B.E. in Mechanical Engineering

May 2019 – May 2023

Relevant Coursework: Programming in C, Joy of Computing Using Python, Machine Learning, Image Processing, Advanced Statistical Methods, Engineering Mathematics, Computational Advanced Numerical Methods

Academic Focus: Systems Programming, Distributed Systems, Machine Learning, Numerical Computing

EXPERIENCE

PhonePe, Bangalore

Site Reliability Engineer

Jul 2023 - Present

- Owned reliability and day-2 operations for 50+ production distributed systems, including GitLab CI/CD-driven NGINX configuration deployments, ensuring safe rollouts, fast rollback, and predictable traffic behavior.
- Automated critical database operations (schema changes, backups, query automation) with Python, MySQL, and Bash, while providing 24/7 environment support and leveraging monitoring tools like Grafana, Prometheus, and InfluxDB.
- Worked with Infrastructure-as-Code (Terraform) to provision and manage cloud resources in collaboration with platform teams, enabling repeatable deployments across environments.
- Served as SPOC for TinyProxy infrastructure integrating HAProxy and Apache Traffic Server, ensuring reliable traffic management, performance tuning, and custom whitelisting/blacklisting configuration testing.
- Defined and tracked service SLIs/SLOs, led incident response and postmortems, and drove root cause fixes that reduced MTTR by approx. 25% and prevented repeat production incidents.

PhonePe, Bangalore

Site Reliability Engineer Intern

Feb 2023 - Jun 2023

- Developed and deployed a Django REST API for on-call roster management with a scalable backend using a multi-master MariaDB Galera cluster and NGINX servers with Keepalived for high availability.
- Designed and maintained end-to-end observability using Prometheus and Grafana, covering metrics, alerts, and dashboards for proactive incident detection.

Airbus, Bangalore

Software Development Engineer Intern

May 2022 - Jun 2022

- Contributed to the development of integration models utilizing data structures in C to enhance system performance by 25%.
- Implemented custom network protocols using skip lists and conducted performance analysis to reduce request response times to $O(\log n)$ from $O(n)$.

PROJECTS

- HostsDB - An Infrastructure Inventory Tool:** Designed and implemented HostsDB, a monitoring tool for gathering and visualizing system-level data across all hosts. Leveraged Osquery, Python, Elasticsearch and Kibana to track system metrics, package versions, and hardware details.
- Evicta - Semantic Inference Cache Engine (Open Source):** Designed a high-performance inference cache that enables deterministic semantic reuse of LLM responses using intent normalization, subject-aware routing, TTL freshness control, and LRU eviction to reduce redundant model calls. Built a microsecond-level hot path for exact and intent cache hits, with extensible inspection and safety gating for controlled reuse.

SKILLS

Languages: Python, Bash, SQL, C/C++

Platforms & Tools: Linux (Ubuntu, Debian, RHEL), Django (REST API development), MariaDB (Galera), MySQL, Elasticsearch, Aerospike, RabbitMQ, Microsoft Azure, On-premise Cloud, Terraform, Ansible, SaltStack, NGINX, HAProxy, Apache Traffic Server, Keepalived, Prometheus, Grafana, Kibana, Docker, Kubernetes (foundational)